

Boolean Time-Slice Scheduling for Cooperative Multi-Target Tracking With Phased-Array Sensors

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Abstract—This paper studies cooperative multi-target tracking (CMTT) with phased-array sensors and develops a Bayesian Cramér–Rao lower bound (BCRLB)-driven Boolean time-slice scheduling (BTSS) method that reduces sensing-time usage while maintaining CMTT performance. An equal-interval time-slicing strategy is adopted, and a Boolean coding representation is constructed to describe per-slice phased-array sensor–target assignments in a unified form. Under this representation, key scheduling constraints—including per-sensor mutual exclusion and a revisit requirement—are expressed linearly. A schedule-dependent predicted Bayesian information matrix (BIM) recursion is established and combined with the BCRLB to define a worst-case tracking metric across targets, explicitly capturing the geometry- and dwell-time-dependent information contribution of phased-array measurements. The resulting Boolean optimization jointly determines the sensing sequence and sensor–target assignments over the horizon, aiming at time-efficient CMTT under performance guarantees. Results over Monte Carlo trials show that the proposed method reduces sensing time usage while maintaining the prescribed CMTT performance level.

Keywords—*phased-array sensors, cooperative multi-target tracking, time-slice scheduling, Boolean coding, integer optimization*

I. INTRODUCTION

CMTT with phased-array sensors is a core capability for airborne sensing teams operating under stringent time constraints. In such systems, phased-array sensing enables agile beam steering and flexible dwell-time control, and the achievable tracking quality is tightly coupled to how sensing time is scheduled across targets and platforms because measurement information depends on both dwell time and sensor–target geometry [1]–[5]. As the number of targets grows and revisit requirements become stringent, scheduling decisions must be made at the level of time slices [6], jointly determining revisit timing and phased-array sensor–target assignments across the horizon.

A large body of work has investigated dwell-time allocation and sequencing for multi-target tracking with phased-array radar and radar networks, demonstrating substantial gains in estimation accuracy and resource utilization [2], [7]–[13]. However, many existing formulations maximize tracking performance under a fixed sensing-time budget, whereas the present work takes a different perspective by minimizing sensing-time usage under prescribed tracking-performance guarantees. In cooperative settings, this performance-constrained time-saving objective is practically meaningful, yet it introduces a more tightly coupled scheduling problem. This setting is challenging because, within each short time slice, each phased-array sensor can illuminate at most one target and each target must be revisited within a bounded interval to maintain track continuity [3], [10], [11]. Together with geometry-dependent information contributions, these requirements lead to high-dimensional Boolean scheduling problems that are difficult to model without losing physical interpretability [5], [10], [12].

This paper develops a BCRLB-driven Boolean time-slice scheduling framework for CMTT with phased-array sensors, explicitly targeting performance-constrained time minimization rather than tracking maximization under a fixed sensing-time budget. The main contributions are:

1) Boolean time-slice coding representation: a unified Boolean schedule is constructed to represent phased-array sensor–target assignments over equal-length slices, enabling linear constraint expressions for mutual exclusion [14] and revisit requirements.

2) Performance-constrained time-minimization optimization: a schedule-dependent predicted BIM [15] recursion is derived and combined with the BCRLB [16] to quantify CMTT performance, consistent with the phased-array measurement model [7]. Based on this metric, the time-slice scheduling problem is formulated as a Boolean integer optimization, jointly determining the sensing sequence and sensor–target assignments while reducing sensing-time usage under a prescribed CMTT performance requirement.

II. SYSTEM MODEL

Consider a team of N airborne sensing nodes cooperatively tracking Q maneuvering targets in three-

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dimensional space. Each sensing node is equipped with an onboard phased-array sensor that provides agile time-on-target control. Time is indexed by frames with interval T . The position vector of target q at frame k is $\mathbf{x}_{q,k}^p = [x_{q,k}, y_{q,k}, z_{q,k}]^T$. The state vector is $\mathbf{x}_{q,k} = [x_{q,k}, \dot{x}_{q,k}, y_{q,k}, \dot{y}_{q,k}, z_{q,k}, \dot{z}_{q,k}]^T$, where $\dot{x}_{q,k}$, $\dot{y}_{q,k}$, and $\dot{z}_{q,k}$ denote the velocity components of target q along the x -, y -, and z -axes at frame k , respectively. The position vector of airborne sensing node n at frame k is $\mathbf{p}_{n,k} = [x_{n,k}, y_{n,k}, z_{n,k}]^T$.

A. Target Motion Model

For all targets, a nearly-constant-velocity model is adopted [17]:

$$\mathbf{x}_{q,k} = \mathbf{F}\mathbf{x}_{q,k-1} + \mathbf{u}_{q,k-1} \quad (1)$$

where $\mathbf{F}$ is the state transition matrix, $\mathbf{u}_{q,k-1}$ is zero-mean Gaussian process noise with covariance matrix $\mathbf{Q}_{q,k-1}$. $\mathbf{F}$ and $\mathbf{Q}_{q,k-1}$ are related to the time interval T between consecutive frames [17].

B. Phased-Array Measurement Model

Each sensing node provides range and azimuth measurements of the target. At frame k , the onboard radar n 's measurement $\mathbf{z}_{n,q,k}$ for target q is expressed as [7]:

$$\mathbf{z}_{n,q,k} = \mathbf{h}(\mathbf{x}_{q,k}^p, \mathbf{p}_{n,k}) + \mathbf{w}_{n,q,k} \quad (2)$$

where $\mathbf{h}(\mathbf{x}_{q,k}^p, \mathbf{p}_{n,k})$ is the measurement function, $\mathbf{w}_{n,q,k}$ is zero-mean Gaussian measurement noise with covariance matrix $\Sigma_{n,q,k}$, which is a diagonal matrix whose diagonal elements determine the mean squared error of the measurement, being inversely proportional to the sensor's dwell time on the target and proportional to the square of the distance between the sensor and the target [7]. This dwell-time- and geometry-dependent measurement quality motivates slice-level scheduling to achieve time-efficient CMTT under prescribed performance requirements.

III. BTSS METHOD

To improve time efficiency under CMTT performance guarantees for an airborne sensing team equipped with phased-array sensors, scheduling must jointly determine dwell-time allocation, slice-level revisit timing, and sensor-target assignments. In phased-array tracking, measurement information depends strongly on dwell time and sensor-target geometry, leading to time-varying information gains. The scheduler therefore needs to enforce operational constraints, maintain timely revisits, and link decisions to a CMTT performance metric, which yields a coupled discrete optimization problem.

To address this challenge, a time-slicing strategy [6] is adopted to structure the sensing-time axis and to express scheduling decisions at a unified slice level. As illustrated in Fig. 1, the sensing time is partitioned into successive slices, and each phased-array sensor is assigned to at most one target per slice. A Boolean coding representation is then

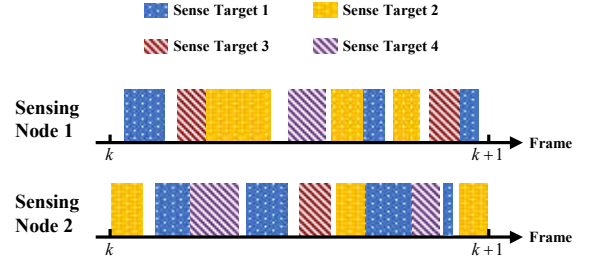


Fig. 1. Time-slice scheduling based on a time-slicing strategy

constructed to describe cross-slice assignments and to yield linear scheduling constraints. Under this representation, a schedule-dependent predicted BIM recursion is derived and combined with the BCRLB to define a worst-case CMTT metric. The resulting Boolean scheduling problem is addressed using a discrete optimization procedure operating directly on the Boolean encoding.

A. Coding Representation and Linearized Constraints

To further simplify the above scheduling problem and to facilitate tractable modeling at the slice level, an equal-interval time-slicing strategy is adopted within each frame interval. Specifically, the time segment from frame k to frame $k+1$ is evenly partitioned into L non-overlapping time slices of identical duration $\tau = T/L$, indexed by $l \in \{1, \dots, L\}$. Define the Boolean coding variable:

$$b_{n,q,l,k} \in \{0,1\} \quad (3)$$

where $b_{n,q,l,k} = 1$ indicates that phased-array sensor n is scheduled to sense target q during time slice l ; otherwise $b_{n,q,l,k} = 0$. The set $\{b_{n,q,l,k}\}$ forms a coding representation of the schedule, and it unifies the key decisions of sensing sequence over time and sensor-target assignment per slice into a single discrete object.

Under the Boolean coding representation, the main operational constraints can be expressed linearly:

1) Mutual exclusion: each phased-array sensor can be assigned to at most one target per slice:

$$\sum_{q=1}^Q b_{n,q,l,k} \leq 1, \quad \forall n, l, k \quad (4)$$

2) Revisit requirement for track maintenance: each target must be revisited at least once within any L' consecutive slices, preventing excessive gaps that would cause rapid uncertainty growth:

$$\sum_{l'=0}^{L'} \sum_{n=1}^N b_{n,q,l+l',k} \geq 1, \quad \forall q, l \in \{1, \dots, L-L'\}, k \quad (5)$$

B. Schedule-Dependent CMTT Performance Metric

The Boolean schedule $\{b_{n,q,l,k}\}$ has a nonlinear mapping relationship with CMTT performance, which is pre-evaluated through the predicted BIM and the corresponding BCRLB, since the slice-level sensing decisions enter the metric directly through the measurement-information term.

Specifically, the Boolean variable $b_{n,q,l,k}$ acts as a switch variable for the measurement information module coupled into the airborne team's predicted BIM [15]:

$$\bar{\mathbf{J}}_{q,l,k} = \left(\bar{\mathbf{F}}(\bar{\mathbf{J}}_{q,l-1,k})^{-1} \bar{\mathbf{F}}^T + \bar{\mathbf{Q}}_c \right)^{-1} + \sum_{n=1}^N b_{n,q,l,k} \bar{\mathbf{H}}_{n,q,l,k}^T \bar{\Sigma}_{n,q,l,k}^{-1} \bar{\mathbf{H}}_{n,q,l,k} \Big|_{\bar{\mathbf{x}}_{q,l,k}} \quad (6)$$

where $\bar{\mathbf{J}}_{q,0,k} = \mathbf{J}_{q,k}$ is the known BIM of the airborne team for target q at frame k , $\bar{\mathbf{J}}_{q,l,k}$ is the predicted BIM at the end frame of time slice l for target q , $\bar{\mathbf{F}}$ and $\bar{\mathbf{Q}}_c$ are the state transition matrix and process noise covariance matrix for the target based on the time slice length τ [17], $\bar{\mathbf{x}}_{q,l,k}$ is the predicted state value of target q at the end frame of time slice l [18], $\bar{\Sigma}_{n,q,l,k}$ is the measurement noise covariance matrix with τ as the beam dwell time [7], $\bar{\mathbf{H}}_{n,q,l,k} = \nabla_{\bar{\mathbf{x}}_{q,l,k}} \mathbf{h}(\bar{\mathbf{x}}_{q,l,k}^p, \mathbf{p}_{n,l,k})$ is the Jacobian matrix of the measurement function [7], where ∇ denotes the partial derivative operator, $\bar{\mathbf{x}}_{q,l,k}^p$ is the predicted position of target q at the end frame of time slice l [18], $\mathbf{p}_{n,l,k}$ is the position vector of sensing node n at the end frame of time slice l . The known BIM of the airborne team for target q at frame $k+1$ is set to the predicted BIM at the end frame of time slice L , i.e., $\mathbf{J}_{q,k+1} = \bar{\mathbf{J}}_{q,L,k}$.

The maximum tracking error among the targets is used as the airborne team's CMTT performance metric. The predicted BCRLB is obtained from the inverse matrix of the predicted BIM, and the team-level CMTT performance metric is expressed as [18]:

$$\mathbb{F}(\{b_{n,q,l,k}\}) = \max_q \sqrt{\text{Tr} \left(\Lambda^T \left(\mathbf{J}_{q,k+1}(\{b_{n,q,l,k}\}) \right)^{-1} \Lambda \right)} \quad (7)$$

where Λ is a normalization matrix [18].

C. Boolean Scheduling Optimization Model

Under the above metric (7), the BTSS problem is formulated as a constrained integer optimization that minimizes sensing-time usage while meeting the required CMTT performance:

$$\begin{aligned} \min_{\{b_{n,q,l,k}\}} & \sum_{n=1}^N \sum_{q=1}^Q \sum_{l=1}^L b_{n,q,l,k} \\ \text{s.t.} & \begin{cases} \sum_{q=1}^Q b_{n,q,l,k} \leq 1, \quad \forall n, l \\ \sum_{l'=0}^{L'} \sum_{n=1}^N b_{n,q,l+l',k} \geq 1, \quad \forall q, l \in \{1, \dots, L-L'\} \\ \mathbb{F}(\{b_{n,q,l,k}\}) \leq \mathbb{F}_k \\ b_{n,q,l,k} \in \{0, 1\}, \quad \forall n, q, l \end{cases} \end{aligned} \quad (8)$$

where $\mathbb{F}_k$ is the preset tracking accuracy requirement value.

The problem (8) is a high-dimensional integer optimization with intrinsically discrete decision variables,

and its dimension scales with the numbers of sensors, targets, and time slices. Continuous relaxations can undermine the semantics of key scheduling constraints, whereas a genetic algorithm (GA) [19] directly operates on Boolean structures and scales well to large combinatorial search spaces [20], [21]. Accordingly, GA is employed.

IV. SIMULATION RESULTS AND ANALYSIS

A. Simulation Setup

To evaluate the proposed BTSS method under a representative phased-array CMTT setting, the key simulation parameters are specified as follows. The frame interval is $T = 1$ s and the scheduling horizon is 40 s. Each frame interval is evenly divided into $L = 10$ time slices. The airborne team contains $N = 2$ phased-array sensing nodes initialized respectively at $[6.5, 5, 2]^T$ km and $[13.5, 5, 2]^T$ km that both move in straight lines at constant speeds $[0, 150, 0]^T$ m/s, and cooperatively tracks $Q = 3$ targets following the near-constant-velocity motion model in Section II.A, with their initial positions and velocities listed in TABLE I. The revisit window is set to $L' = 5$. The simulation geometry of sensors and targets is shown in Fig. 2.

The tracking accuracy requirement is specified as a frame-dependent threshold sequence $\{\mathbb{F}_k\}$, consistent with the per-frame constraint used in the optimization model. This threshold curve is calibrated using the cyclic scheduling (CS) baseline, which follows a fixed round-robin illumination pattern with one target assigned to each phased-array sensor per slice under the same mutual-exclusion and revisit settings. Specifically, CS is executed for 100 Monte Carlo trials using the full horizon, and $\mathbb{F}_k$ is set to the 10th-percentile value of the metric in (7) across trials, which yields a conservative yet practically attainable time-varying requirement under full sensing-time usage.

B. Experimental Results and Analysis

Across 100 Monte Carlo trials, the proposed BTSS method attains an average threshold-satisfaction rate of 91.56%; all trials exhibit convergent tracking-error trajectories, and the maximum exceedance over all trials and frames is only 0.5526 m. All generated time allocation results are verified to satisfy constraints (4) and (5) throughout the horizon. Fig. 3 plots the resulting average maximum target tracking error, computed by the same metric as in (7), together with the tracking accuracy requirement curve (TAR). The BTSS curve remains tightly aligned with TAR over the horizon, which is consistent with the objective of reducing sensing-time usage while

TABLE I. TARGET STATE SETTINGS

Parameter	Value
Target 1's initial position	$[10.5, 30, 2]^T$ km
Target 1's initial velocity	$[-90, -320, 0]^T$ m/s
Target 2's initial position	$[13, 25, 2]^T$ km
Target 2's initial velocity	$[40, -110, 0]^T$ m/s
Target 3's initial position	$[4, 22, 2]^T$ km
Target 3's initial velocity	$[10, -120, 0]^T$ m/s

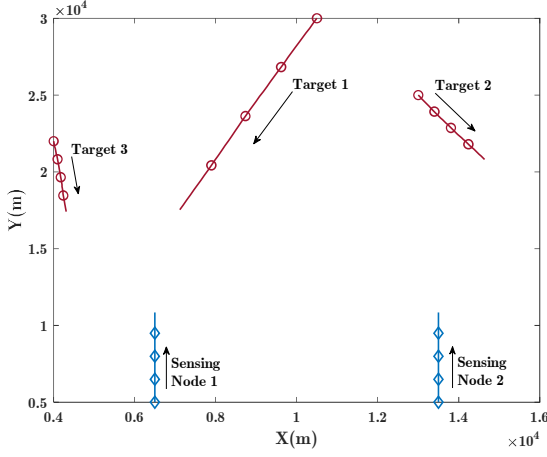


Fig. 2. Simulation geometry of phased-array sensors and targets

maintaining CMTT performance at the required level. Fig. 4 reports the per-frame average sensing-time saving rate, obtained by averaging the saving rate at each frame over the 100 trials, and yields an overall mean saving of 35.50% over the mission. Jointly interpreting Fig. 3 and Fig. 4 shows that the saving rate drops in frames 20–26 and reaches its minimum around frames 20–21, where the CMTT burden is highest—reflected by the peak of the tracking-error curve in Fig. 3—thus requiring substantially more, sometimes nearly full sensing time to stabilize tracking performance.

To further interpret the trends observed in Fig. 3 and Fig. 4, Fig. 5 presents the slice-level sensor–target time allocation result produced by BTSS for one representative run selected from the trials, where a clear geometry-driven allocation behavior is observed. In the early stage, the CMTT burden is highest and the tracking-error convergence pressure is strongest under the time-saving objective, so BTSS allocates a dominant share of slices to Target 1, which is farthest from the sensing formation and provides the weakest instantaneous measurement information. The remaining slices are assigned to the closer targets following proximity, with Sensing Node 1 mainly serving Target 3 and Sensing Node 2 mainly serving Target 2. This allocation is consistent with phased-array tracking, where distant targets typically require more dwell time to prevent them from dominating the worst-case tracking error. As the task proceeds, the tracking errors progressively stabilize, especially for the fast-moving Target 1 as it approaches the formation and its observability improves. Accordingly, both sensing nodes reduce the slices assigned to Target 1 and redistribute them to Targets 2 and 3 to further control the worst-case tracking error. Target 2 usually receives a larger share than Target 3 because it remains relatively farther and is more likely to become the worst-case term if under-served.

V. CONCLUSION

This paper develops a BTSS method for CMTT with phased-array sensors, targeting sensing-time minimization subject to CMTT performance requirements. An equal-interval time-slicing strategy is employed to structure slice-level decisions, and a Boolean coding representation is introduced to describe sensor–target assignments in a unified and tractable form. With this representation, key scheduling constraints—such as per-sensor mutual exclusion and revisit requirements—admit linear expressions. Moreover, a schedule-dependent predicted

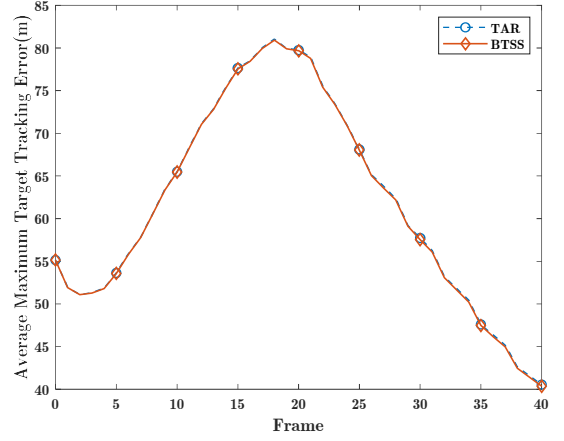


Fig. 3. Average maximum target tracking error curves

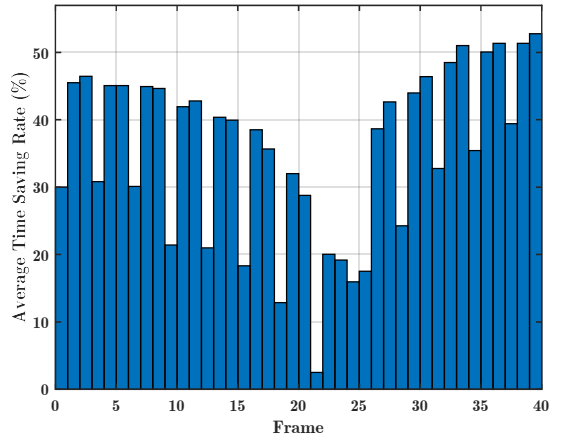


Fig. 4. Average time saving rate

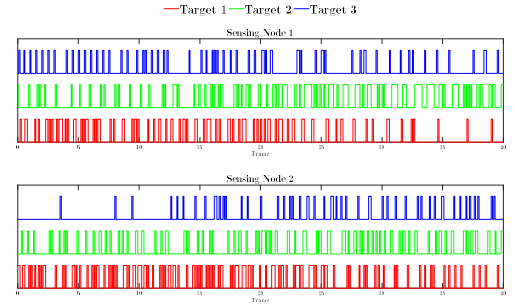


Fig. 5. Slice-level sensor–target time allocation result of BTSS method

BIM recursion is derived and combined with the BCRLB to construct a worst-case tracking metric across targets, thereby establishing a principled link between phased-array sensing schedules and cooperative tracking quality. These elements together yield a BTSS optimization formulation that supports joint planning of sensing sequencing and sensor–target assignments while explicitly enforcing tracking requirements, enabling time-efficient CMTT by reducing sensing-time usage while maintaining the required performance level.

Future work will focus on improving the computational efficiency of Boolean scheduling and extending the framework toward joint beam–time scheduling and more complex cooperative tracking scenarios.

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